# DUA SHAH

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## PROFESSIONAL SUMMARY

Computer Science graduate (BSCS, CGPA 3.63, merit scholarship) with hands-on experience building full-stack web applications and machine learning systems. Developed an AI-powered financial management platform as a final year project, plus blockchain verification, fraud detection, and predictive modeling projects using Python, FastAPI, React.js, Node.js, and scikit-learn. Seeking an entry-level software engineering, data science, or machine learning role.

## EDUCATION

**Bachelor of Science in Computer Science (BSCS)** | Iqra University, Karachi | 2022 – 2026
CGPA: 3.63 / 4.00 | Awarded merit scholarship

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, Java, SQL, Solidity, HTML, CSS
**Web Development:** React.js, Node.js, Express.js, FastAPI, Streamlit, Tailwind CSS, REST APIs
**Machine Learning & Data:** scikit-learn, Pandas, XGBoost, SMOTE, Matplotlib, Logistic Regression, SVM, Random Forest, Artificial Neural Networks, Data Preprocessing
**Databases:** MariaDB, MySQL, SQLAlchemy
**Blockchain:** Ethereum, Solidity, Hardhat, Web3.py, JSON-RPC, Solana
**Other:** OCR (EasyOCR), Data Structures & Algorithms, Git, Object-Oriented Programming

## PROJECTS

**SmartFinance — AI-Powered Financial Management System** (Final Year Project) | React.js, Node.js, Express.js, MariaDB, Tailwind CSS, REST API

- Developed a full-stack web app for small business owners to manage multiple businesses, track income and expenses, monitor profit/loss, and generate financial reports.
- Implemented transaction categorization, spending pattern analysis, budget alerts, and AI-based expense prediction to surface actionable insights.
- Built an admin dashboard and responsive light/dark-mode UI, backed by a REST API and relational database.

**ChainCred — Blockchain-Based Degree Attestation System with ML Risk Assessment** | Python, FastAPI, Solidity, scikit-learn

- Designed a platform that verifies academic credentials and records a cryptographic hash of each degree on a private Ethereum blockchain via a Solidity smart contract deployed with Hardhat.
- Implemented an automated document verification pipeline using EasyOCR to validate CNIC and marksheet data, enforcing eligibility rules before attestation.
- Trained a supervised classifier (scikit-learn) scoring application genuineness from document quality, OCR confidence, and eligibility signals to support reviewer decisions.
- Built a FastAPI/SQLAlchemy backend with OTP authentication and an admin review workflow, plus PDF certificates embedding the on-chain transaction reference.

**ChainGuard — Multi-Chain Blockchain Fraud Detection Tool** | Python, Streamlit, JSON-RPC APIs

- Built a Streamlit app analyzing Solana and Ethereum transactions in real time via public JSON-RPC nodes, removing any dependency on paid third-party APIs.
- Designed a multi-factor risk-scoring engine (6+ fraud indicators: transaction failure, fee anomalies, large-value transfers, contract complexity) classifying transactions as low/medium/high risk, with separate pipelines for Solana and Ethereum.

**Diabetes Prediction & Data Mining System** | Python, Pandas, scikit-learn, XGBoost, SMOTE

- Performed data mining and preprocessing on physiological health data (blood glucose, blood pressure, heart rate, body temperature, SpO2, age) to identify factors associated with diabetes.
- Compared Logistic Regression, SVM, Random Forest, XGBoost, and ANN classifiers, applied SMOTE to address class imbalance, and evaluated models using accuracy, precision, recall, F1-score, and confusion matrix.

**University Event & Resource Allocation System** | Java, Data Structures & Algorithms

- Created a command-line application for scheduling university events using linked lists, queues, stacks, hash maps, and a conflict graph, with search and sorting algorithms and undo functionality.

## EXPERIENCE

**Customer Sales Representative** (with Quality Control responsibilities) | Star Dynamics Solutions | 2022

- Provided customer support via phone, email, and chat, and maintained accurate CRM records for follow-up.
- Met or exceeded customer satisfaction and response-time targets while supporting quality control checks.

## CERTIFICATIONS

Data Science Job Simulation — British Airways (Forage), 2026 | Data Analyst 101, 2026 | Intro to Web Development with ChatGPT, 2026 | Fiverr Freelancing, 2023 | English Spoken Language, 2023